David Aretz

PhD student in Mathematics

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Research interests

Areas: Higher differential geometry; supersymmetric functorial field theories; homotopy theory and K-theory; noncommutative geometry.

Education

PhD **University of Bonn and Max Planck Institute for Mathematics, Bonn**, Mathematics — Apr 2023 – 2027
- Supervisors: Christian Blohmann and Peter Teichner.
- Provisional thesis title: *Classifying spaces of supersymmetric field theories*

MSc **University of Bonn**, Mathematics — Oct 2020 – Jan 2023
- Final grade 1.0, with distinction. Supervisor: Christian Blohmann.
- Thesis: *Cohomological Extension of Groupoid Convolution*

BSc **University of Bonn**, Mathematics — Oct 2017 – Aug 2020
- Final grade 1.0, with distinction. Secondary subject: Astrophysics. Supervisor: Christian Blohmann.
- Thesis: *The Noncommutative Geometry of Symplectic Singularities*

Experience

University of Bonn, Tutor, Higher Category Theory (lecturer Merlin Christ) — Oct 2026 – present, 1 month

University of Bonn, Mathematical Institute, Workshops for new tutors — 2022 – 2026, 4 years
Leading one of the one-day workshops for new tutors (2022, 2024, 2025, 2026), with Regula Krapf.

University of Bonn, Mathematical Institute, Tutor — Sept 2018 – Mar 2023, 4 years 7 months
- Geometry and Analysis on Manifolds (winter 2021/22, 2022/23)
- Riemannian Geometry (summer 2022)
- Analysis I (winter 2020/21), Analysis II (summer 2021), Analysis III and revision course (winter 2019/20)
- Complex Analysis (summer 2020)
- Mathematics precourses (2018, 2019)

Parental leave

- June–July 2024

Languages

Languages
English, German, Spanish, Latin, Arabic (beginner)